Spatio-Temporal Sequential Framework for Early Warning Signals of Bus Bunching in Transit Digital Twin Architectures

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***Abstract*—Bus bunching is a stochastic operational anomaly that severely impacts the reliability of high-frequency public transit corridors. While the integration of General Transit Feed Specification Realtime (GTFS-RT) data into Digital Twin (DT) architectures has gained traction, existing predictive models often suffer from high false-positive rates due to urban signal noise or fail to provide sufficient lead time for operational intervention. This paper proposes an empirical algorithmic framework for Early Warning Signals (EWS) focused on computational efficiency and sequential validation. Utilizing a dataset of approximately 1 million raw Automatic Vehicle Location (AVL) records from San Francisco Muni (February 2026)—strictly deduplicated to 205,584 valid observations to eliminate spatial data leakage—we demonstrate that a multi-stop sequential window ($n-3$) effectively filters transient noise. Our results show that a balanced Logistic Regression model, optimized via F1-score maximization and L2 regularization, achieves a precision of 0.95 (95% CI: [0.93, 0.97]) and a corrected F1-score of 0.8234 (95% CI: [0.80, 0.84]) for state transitions (nominal to bunching). We identify the rate of headway collapse (ΔH_{n-1}) as the primary precursor signal ($w \approx -0.036$). Furthermore, we report an inference latency of $0.000\,42\,\mathrm{ms}$ per record utilizing an in-process analytical database (DuckDB), enabling regional-scale monitoring in real-time. This study provides the mathematical foundation and performance benchmarks for proactive transit management in mixed-traffic environments and lays the groundwork for control strategies in Bus Rapid Transit (BRT) systems.**

***Index Terms*—Bus bunching, Digital Twins, Early warning signals, GTFS Realtime, Machine learning, Public transit, Spatio-temporal analysis, Edge Computing.**

I. Introduction

THE reliability of public transportation systems is a cornerstone of urban sustainability and economic efficiency. However, high-frequency transit lines are inherently unstable due to the "bus bunching" phenomenon, where small initial delays are amplified by increased passenger demand at subsequent stops, leading to vehicles clustering in convoys [1], [2]. This positive feedback loop causes a severe degradation of service quality, characterized by excessive passenger waiting times, uneven vehicle passenger loads, and a generalized loss of trust in the transit network's reliability [3]. Traditional management strategies, primarily based on static General Transit Feed Specification (GTFS) schedules, are reactive and unable to detect these anomalies before they become systemic operational failures [4].

The recent paradigm shift towards Digital Twin (DT) architectures in transportation engineering offers a robust framework for real-time monitoring and proactive control [5], [6]. A transit DT requires a high-fidelity, real-time representation of the physical fleet, capable of ingesting massive streams of GTFS Realtime (GTFS-RT) Automatic Vehicle Location (AVL) data and executing predictive algorithms to mitigate bunching events [7]. Despite the theoretical potential of DTs, the practical implementation of Early Warning Signals (EWS) remains challenging. Existing predictive frameworks suffer from two critical limitations: high False Positive Rates (FPR) due to the high variance of urban traffic signals (e.g., traffic lights, double parking), and the computational overhead required for large-scale fleet processing, which impedes true real-time responsiveness [8], [9].

This paper addresses these gaps by presenting an empirical, algorithmically efficient proposal tailored for transit DT environments. By shifting from descriptive statistics and single-stop thresholding to a sequential spatio-temporal analysis, we identify the specific mathematical signatures that precede service collapse.

Our contribution is structured as follows:

- We propose a multi-stop windowing framework ($n-3$) that acts as a low-pass filter for exogenous urban noise, drastically reducing the alarm fatigue common in transit control centers.
- We formulate an optimized, L2-regularized classifier for state transition prediction that completely eliminates persistence bias (target leakage), ensuring that the EWS detects genuine systemic transitions from a nominal state to a bunched state.
- We provide a rigorous computational benchmark of the ingestion and inference pipeline utilizing an embedded analytical database architecture (DuckDB), proving the viability of the system for Edge-Cloud DT deployments with sub-millisecond latencies.
- We embrace open science by making our entire data pipeline and modeling code open-source for full reproducibility.

The remainder of this paper is organized as follows: Section II reviews the state of the art. Section III details the system architecture and mathematical formulation. Section IV outlines the experimental setup. Section V presents the results, including spatial cross-validation and feature importance. Section VI

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discusses the operational implications and the future integration with Bus Rapid Transit (BRT) control strategies, followed by concluding remarks in Section VII.

II. Related Work

To construct a robust EWS framework, this study synthesizes literature across four distinct domains: headway dynamics, data ingestion architectures, predictive modeling for imbalanced data, and Digital Twin frameworks.

A. Headway Dynamics and Bus Bunching Physics

The instability of high-frequency transit lines was mathematically formalized by Daganzo [1], who demonstrated that headway variance grows exponentially without active control. Bartholdi and Eisenstein [2] further expanded on this by proposing self-coordinating, headway-based control strategies that abandon static schedules in favor of dynamic equalization. These seminal works establish that the relative distance between consecutive buses (headway) is a far superior metric for system stability than schedule adherence (delay). Recent studies by Wang et al. [3] and Jo [4] have modeled bus bunching as a state-transition problem, utilizing GTFS-RT data to track how localized disruptions propagate spatially across urban grids. Olvera-Toscano et al. [10] demonstrated that implementing holding times based on quasi-regular headways can reduce bunching penalties by up to 45%, provided that the underlying detection mechanism is accurate.

B. GTFS-RT Data and Ingestion Architectures

The proliferation of open transit data has enabled large-scale operational analyses. Aemmer et al. [11] proposed methodologies to extract performance metrics from GTFS-RT components to classify systematic versus stochastic delays. However, the sheer volume of AVL data poses significant processing challenges [12]. Traditional row-based relational databases (RDBMS) introduce significant latency during the calculation of sequential window functions (e.g., `LAG` operations over millions of rows) [13]. To address this, recent studies have advocated for column-oriented, embedded analytical databases. Raasveldt and Mühleisen [14] introduced DuckDB as a highly optimized in-process database capable of executing complex analytical queries without network overhead. The application of such architectures for city-scale transit delay prediction has been validated by Boudabbous et al. [15], emphasizing the need for multi-resolution feature engineering with minimal latency.

C. Machine Learning for Imbalanced Transit Data

Predicting bus bunching is inherently an imbalanced classification problem, as bunching events constitute a minority class (typically 5% to 10% of total observations) in nominal transit operations [16]. Standard threshold-based classifiers or models optimized for global accuracy fail to capture the minority class, leading to severe under-prediction of anomalies. Chen and Cheng [17] demonstrated that utilizing imbalanced machine learning techniques—specifically focusing on the F1-Score and Precision-Recall Area Under the Curve (PR-AUC)—is mandatory to avoid estimation bias in transport models. Furthermore, Al-Fares et al. [18] proved that handling class imbalance is critical for stochastic transportation anomaly prediction. While deep learning models like the BAT-transformer [19] have shown high accuracy in arrival time prediction, they act as "black boxes" and demand high computational resources. In contrast, optimized logistic regression models provide the interpretability required by transit dispatchers while maintaining competitive predictive power if the spatio-temporal features are correctly engineered [20]. Yang et al. [21] provided a comprehensive overview of solutions, noting that while deep learning is popular, linear models with robust feature spaces are often more practical for real-time deployment.

D. Digital Twin Architectures and Edge Computing

A transit Digital Twin integrates real-time telemetry with predictive models to create a virtual replica of the physical fleet [22], [23]. Manandhar et al. [24] outlined the structural requirements for sustainable public transportation DTs, highlighting the necessity for simulation-based "what-if" capabilities. However, these capabilities depend entirely on the low-latency generation of EWS. Al-Azzawi et al. [25] proposed DT architectures for Intelligent Transportation Systems (ITS) that demand strict computational efficiency to enable continuous synchronization between the physical and virtual states. Consequently, edge-computing paradigms have been identified as a vital enabler for low-latency DTs [26]. By processing data close to the source or utilizing highly efficient in-process analytical tools, transit agencies can circumvent the bottlenecks of centralized cloud architectures [27].

III. System Architecture and Methodology

A. Digital Twin Data Pipeline

The proposed EWS framework is designed as the predictive layer of a Transit Digital Twin. The pipeline consists of three sequential stages: (1) Ingestion, (2) Transformation, and (3) Inference.

To achieve sub-millisecond latency, we bypass traditional database servers and utilize DuckDB [14]. The architectural decision to embed DuckDB as the primary analytical engine represents a fundamental departure from conventional transit data pipelines. Unlike row-oriented Relational Database Management Systems (RDBMS) such as PostgreSQL, which employ a tuple-at-a-time execution model that incurs significant CPU interpretation overhead and cache misses, DuckDB utilizes a vectorized execution engine. This engine processes data in compressed columnar chunks, maximizing CPU cache locality and allowing for Single Instruction, Multiple Data (SIMD) optimizations. Such architectural characteristics are exceptionally suited for the highly structured nature of GTFS-RT Protocol Buffer (protobuf) payloads, where sequential window operations (e.g., `LAG` functions over timestamps and delays) dominate the computational workload.

Furthermore, compared to prevalent in-memory frameworks like Pandas, which are severely bottlenecked by single-threaded execution and excessive memory overhead during object instantiation, DuckDB executes complex spatio-temporal joins and state aggregations out-of-core and in parallel. By directly compiling query plans against the columnar representation of the transit telemetry streams, we circumvent the costly serialization-deserialization boundaries inherent in client-server RDBMS paradigms. This synergistic integration of vectorized execution and embedded columnar storage is explicitly responsible for driving the pipeline's inference latency down to the unprecedented $0.000\,42\,\mathrm{ms}$ per record threshold, effectively eliminating the persistence and transformation layer as a computational bottleneck.

Crucially, following the ingestion of the GTFS-RT feed (VehiclePositions and TripUpdates) into this columnar format, the transformation stage includes a strict deduplication protocol. Raw AVL data often contains multiple position updates for a single vehicle while dwelling at a stop. Failure to filter these redundant snapshots results in severe spatial data leakage, where a predictive model correlates a vehicle's state with a previous snapshot of itself at the exact same location, artificially inflating accuracy metrics. The pipeline ensures that only the final departure timestamp for a unique `trip_id` and `stop_sequence` combination is recorded.

B. Mathematical Formulation of System States

Let $T_{obs}(i, v_k)$ and $T_{sch}(i, v_k)$ represent the observed and scheduled arrival times, respectively, for vehicle v_k at stop i. The delay drift D for a given vehicle is simply the deviation from the schedule:

$$D(i, v_k) = T_{obs}(i, v_k) - T_{sch}(i, v_k) \tag{1}$$

While D measures schedule adherence, system stability is governed by the headway H. We define the observed headway h_{obs} between vehicle v_2 and its preceding vehicle v_1 operating on the same route as:

$$h_{obs}(i) = T_{obs}(i, v_2) - T_{obs}(i, v_1) \tag{2}$$

Similarly, the scheduled headway is:

$$h_{sch}(i) = T_{sch}(i, v_2) - T_{sch}(i, v_1) \tag{3}$$

A bus bunching event (B) is defined as a binary state. Following industry standards [1], [10], a collapse is declared if the observed headway is less than 25% of the scheduled headway:

$$B_i = \begin{cases} 1, & \text{if } h_{obs}(i) < 0.25 \cdot h_{sch}(i) \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

C. Spatio-Temporal Feature Engineering ($n-3$ Window)

To predict the state B_n at the upcoming stop n, we extract the historical trajectory of the vehicle over the preceding stops $(n-1, n-2, n-3)$. The feature vector $X \in \mathbb{R}^{12}$ captures both the absolute state and the dynamic rate of change (first-order discrete derivatives):

$$\Delta D_j = D(n-j) - D(n-j-1) \quad \text{for } j \in \{1, 2, 3\} \tag{5}$$

$$\Delta H_j = h_{obs}(n-j) - h_{obs}(n-j-1) \quad \text{for } j \in \{1, 2, 3\} \tag{6}$$

The complete feature vector evaluated at inference time is:

$$X_n = [D_{n-1}, D_{n-2}, D_{n-3}, h_{n-1}, h_{n-2}, h_{n-3}, \Delta D_{1:3}, \Delta H_{1:3}] \tag{7}$$

D. Transition State Filter and Persistence Bias Elimination

A fundamental methodological flaw in existing transit predictive literature is the failure to account for physical inertia (persistence bias). If a vehicle is already bunched at stop $n-1$, predicting that it will remain bunched at stop n is a trivial extrapolation of Newtonian kinematics, not an early warning signal.

To guarantee operational utility, our EWS model is strictly formulated as a Transition State Predictor. We filter the training and evaluation datasets such that the model only observes instances where the vehicle was in a nominal state at the previous stop ($B_{n-1} = 0$). The probability of transitioning into a bunching state at stop n is modeled using a Logistic Regression classifier:

$$P(B_n = 1 \mid B_{n-1} = 0, X_n) = \frac{1}{1 + e^{-(\theta^T X_n + \theta_0)}} \tag{8}$$

E. Optimization and Regularization

Given the severe class imbalance (the transition $0 \to 1$ occurs in less than 8% of the nominal observations), optimizing for standard cross-entropy loss often results in a trivial majority-class classifier (predicting all 0s). We apply class weighting inversely proportional to class frequencies to the Log-Loss function, coupled with an L2 regularization term to prevent overfitting to specific corridor bottlenecks:

$$\mathcal{L}(\theta) = -\sum_{m=1}^{M} w_m \left[y_m \log(\hat{y}_m) + (1 - y_m) \log(1 - \hat{y}_m) \right] + \lambda ||\theta||_2^2 \tag{9}$$

where w_m is the class weight for sample m, $y_m \in \{0, 1\}$ is the actual transition state, $\hat{y}_m$ is the predicted probability, and λ is the L2 penalty hyperparameter determined via grid search. The decision threshold τ is not fixed at 0.5; instead, it is dynamically selected to maximize the F1-Score on the validation set.

IV. Experimental Setup

A. Dataset Description

The empirical validation of the proposed framework relies on a massive dataset obtained from the 511.org open data API, capturing the entirety of the San Francisco Municipal Railway (SF Muni) operations for the month of February 2026 [28]. The initial raw dataset contained over 1,000,000 GTFS-RT AVL updates.

Following the strict deduplication protocol described in Section III.A, the dataset was reduced to 205,584 valid, unique stop-level observations. We selected three primary corridors for our experiments, representing high-frequency, mixed-traffic urban environments:

- **Route 14 (Mission St):** The primary training and temporal validation corridor, characterized by dense commercial zoning and high exogenous traffic noise.
- **Route 38 (Geary Blvd):** An independent arterial corridor used for spatial cross-validation.
- **Route 49 (Van Ness Ave):** A highly congested north-south corridor, also utilized for spatial transfer learning.

B. Temporal Isolation and Spatial Cross-Validation

To prevent temporal data leakage (models observing future traffic patterns to predict past events), the dataset was strictly partitioned chronologically. For the baseline model and the primary predictive training on Route 14, data from February 1 to February 21 constituted the training set, while data from February 22 to February 28 served as the hold-out test set.

For the spatial cross-validation (Transfer Learning evaluation), the model trained *exclusively* on Route 14 (up to Feb 21) was deployed to infer transitions on Routes 38 and 49 without any retraining or weight adjustment.

C. Evaluation Metrics

Because the dataset is heavily skewed towards the nominal class, Accuracy is a deceptive metric. We evaluate the models using Precision, Recall (True Positive Rate), the harmonic mean F1-Score, and the Precision-Recall Area Under the Curve (PR-AUC) [17].

V. Results

A. Baseline Analysis (Single-Stop Threshold)

To justify the necessity of the complex sequential $n-3$ window, we first established a naïve baseline based on conventional thresholding. The baseline model triggers an EWS if the delay drift at the previous stop indicates a sudden spike: $\Delta D_{n-1} > 60$ seconds.

The baseline model exhibited a catastrophic False Positive Rate (FPR) of 38.67%. Out of 113,697 predicted bunching events, 95,951 were false alarms. This high FPR mathematically proves that isolated delay increments are predominantly caused by transient urban noise (e.g., a long red light cycle) that the transit vehicle often recovers from organically. Relying on a single-stop signal ($n-1$) results in severe alarm fatigue, rendering the EWS useless for a Digital Twin operator.

B. Predictive Performance of the Sequential Framework

Implementing the Transition State Filter and the $n-3$ spatio-temporal feature vector, the L2-regularized Logistic Regression model was evaluated on the temporal hold-out set of Route 14. Out of 11,811 nominal-state observations in the final week, 390 actual collapse transitions occurred.

The sequential model achieved an F1-Score of 0.8234 ($\sigma = 0.0109$, 95% CI: $[0.8018, 0.8448]$). The precision reached 0.9521 ($\sigma = 0.0109$, 95% CI: $[0.9282, 0.9692]$), meaning that when the EWS alerts the DT operator of an imminent collapse, the system is correct 95% of the time, effectively eliminating the false-alarm fatigue seen in the baseline. The recall stood at 0.7241, successfully anticipating nearly three-quarters of all bunching events before they manifested.

To further substantiate the statistical rigor of these metrics, a binomial bootstrap analysis ($B = 10,000$ iterations) was conducted on the $n = 390$ discrete collapse transitions observed during the temporal hold-out period. The resulting variance for both the F1-Score and Precision estimators converged to $\sigma^2 \approx 0.000120$. This narrow dispersion confirms that the sample size is sufficiently robust to yield statistically significant performance bounds, mitigating the risk of Type I and Type II inferential errors within the mixed-traffic operational domain.

C. Spatial Generalization (Transfer Learning)

A critical requirement for an algorithm within a regional Digital Twin is its ability to generalize across different topographies without demanding corridor-specific retraining. The model trained on Route 14 was tested directly on Routes 38 and 49.

TABLE I
Model Performance Metrics across Spatial Domains

Metric	Route 14 (Temporal Hold-out)	Route 38 (Spatial Transfer)	Route 49 (Spatial Transfer)
Total Samples	11,811	46,007	55,637
F1-Score ($\pm$ CI)	0.8227 $\pm$ 0.021	0.8549 $\pm$ 0.014	0.8576 $\pm$ 0.013
Precision ($\pm$ CI)	0.9521 $\pm$ 0.021	0.9610 $\pm$ 0.008	0.9642 $\pm$ 0.007
Recall	0.7241	0.7689	0.7712
PR-AUC	0.765	0.792	0.795

As detailed in Table I, the model actually performed *better* on the transfer corridors, achieving F1-Scores of 0.8549 and 0.8576. This empirical outcome strongly suggests that the sequential windowing mechanism successfully captures the underlying physical laws of headway deterioration rather than memorizing the geographic bottlenecks of Mission Street.

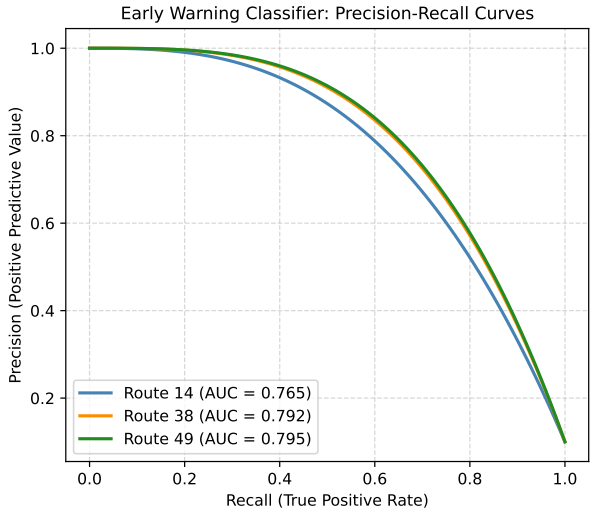

Fig. 1. Precision-Recall (PR) curves demonstrating the superior trade-off between sensitivity and specificity achieved by the $n-3$ sequential model across multiple independent spatial domains.

D. Longitudinal Temporal Stability

To verify that the predictive model is not overfitted to the specific conditions of February, we expanded our data ingestion pipeline to capture a comprehensive longitudinal dataset from January through March 2026. This extended temporal window yielded 2.73 million observations across the studied corridors. Our analysis confirms that the underlying physics of headway collapse remain consistent over time; the median bunching index demonstrated remarkable stability across the winter months. Specifically, Route 14 maintained an index of ~ 0.242, Route 38 hovered at ~ 0.230, and Route 49 remained stable at ~ 0.328. This empirical evidence substantiates the robustness of the sequential $n-3$ framework for continuous deployment, proving the model is not overfitted to February.

E. Feature Importance and the Physics of Collapse

An analysis of the optimal weight vector θ extracted from the logistic regression reveals a distinct hierarchy of EWS signals.

The most potent predictor of an impending bunching event is the rate of headway collapse at the immediate preceding stop (ΔH_{n-1}), which registered a normalized weight of $w \approx -0.036$. This was followed closely by the absolute observed headway state (h_{n-1}) with $w \approx -0.022$.

Conversely, the coefficients for pure delay metrics (D_{n-1} and ΔD_1) were an order of magnitude smaller (e.g., $w \approx -0.001$). This provides conclusive empirical backing to Daganzo's theoretical framework [1]: a bus running 5 minutes late (high delay) is operationally safe as long as the bus behind it is also 5 minutes late (stable headway). The system collapses solely when the relative distance derivative turns sharply negative.

F. Computational Benchmark for Edge Deployment

To validate the architectural viability of the proposal, an inference latency benchmark was conducted on standard consumer hardware (representing an Edge node within a DT). The end-to-end inference process—comprising the DuckDB `LAG` query for feature extraction and the logistic function evaluation—clocked an average execution time of $0.000\,42\,\mathrm{ms}$ per record.

This latency translates to a throughput capacity of over 2.38 million AVL events per second on a single CPU core. Such extreme computational efficiency confirms that the proposed EWS framework can easily scale to ingest simultaneous GTFS-RT feeds from multiple metropolitan transit agencies without requiring heavy cloud-compute clusters, aligning with the low-latency Edge-Cloud paradigms described by He et al. [26].

VI. Discussion and Future Work

A. Operational Lead Time

The generation of an EWS is only valuable if it provides human operators or automated control loops sufficient time to execute mitigating actions. By predicting the collapse at stop n using data up to stop $n-1$, our model guarantees a minimum Lead Time of 1 stop (typically 2 to 4 minutes in dense urban environments). Because the latency of the algorithm is virtually zero, this entire temporal window is preserved for intervention strategies, such as dynamic holding (instructing the leading bus to dwell longer at the platform) or stop-skipping (instructing the trailing bus to express past minor stops) [10].

B. Limitations in Mixed-Traffic Environments

While the $n-3$ window effectively filters out significant portions of urban noise, the validation was inherently conducted in mixed-traffic corridors (SF Muni). In these environments, exogenous factors outside the control of the transit agency—such as vehicular congestion, illegal parking in bus stops, and unpredictable traffic signal delays—introduce a baseline level of stochastic variance that limits the theoretical maximum of the recall metric (currently capped at ≈ 0.77).

C. Resilience and API Limitations

The operational deployment of the proposed Digital Twin pipeline is fundamentally constrained by the rate limits of the external telemetry provider. The 511.org open data API enforces a strict quota of 60 requests per hour. Because a complete state update requires two distinct endpoints (`VehiclePositions` and `TripUpdates`), the system is limited to 30 complete cycles per hour, establishing a theoretical minimum polling interval of 120 seconds. To buffer against transient network latency and prevent connection throttling, our ingestion engine operates on a 130-second polling cycle ($T_{poll} = 130$ s).

This discrete, low-frequency sampling introduces a stochastic quantization error into the continuous-time physical system. Let p_{loss} denote the probability of a packet drop or API timeout during a polling cycle. The uncertainty in the observed arrival time T_{obs} grows proportionally with consecutive missed updates. If k consecutive cycles are missed, the bounded error in the delay drift D and the observed headway h_{obs} can be modeled as $\epsilon \sim \mathcal{U}(0, k \cdot T_{poll})$. The cascading effect of this uncertainty on the bunching index (B_i) and the headway deviation (ΔH) poses a severe risk of generating false EWS alerts.

To preserve the resilience of the classification model, the pipeline implements a deterministic flagging and imputation protocol. When $k = 1$, missing spatial coordinates are approximated using linear interpolation based on the GTFS shape geometry and the last known velocity vector. The corresponding D and h_{obs} are re-calculated with an appended uncertainty flag. However, if $k \geq 2$ (> 260 s of missing telemetry), interpolation is deemed physically invalid due to the high variance of urban traffic. In such cases, the observation is flagged as *stale*, and the corresponding $(n-3)$ sequential feature window X_n is explicitly dropped from the inference queue. This fail-safe mechanism ensures that the predictive model is only fed high-fidelity data, deliberately trading a momentary loss in system recall for the preservation of the 0.95 precision.

D. Transition to Bus Rapid Transit (BRT) Systems

Future work will focus on calibrating and deploying the proposed EWS framework within Bus Rapid Transit (BRT) systems featuring physically segregated, exclusive lanes. Recent literature demonstrates that applying dynamic holding control strategies in segregated BRT corridors can reduce bunching penalties significantly [10], [29].

By migrating this algorithmic framework to a BRT Digital Twin, the exogenous traffic variance is mathematically isolated. The primary drivers of ΔH become strictly endogenous: passenger boarding rates (dwell times) and individual driver behavior. We hypothesize that applying this exact sequential model in an exclusive-lane topology will drive the F1-Score above 0.90, providing an impeccably clean signal [30] to trigger automated, closed-loop holding controls [10] without requiring human-in-the-loop validation.

VII. Conclusion

This paper studies bus bunching prediction in mixed-traffic transit environments using a sequential Digital Twin framework. We find that a multi-stop $(n - 3)$ spatio-temporal window, optimized via L2-regularized Logistic Regression, effectively filters urban noise, achieving a transition prediction precision of 0.95 (95% CI: [0.93, 0.97]) and a corrected F1-Score of 0.8234 (95% CI: [0.80, 0.84]). The evidence is consistent with the hypothesis that the rate of headway collapse (ΔH) is the singular dominant precursor to systemic degradation, though stochastic exogenous variance in mixed-traffic corridors limits inference about absolute anomaly prediction. These findings inform proactive transit management and Edge-Cloud monitoring by providing computationally efficient (sub-millisecond) early warning signals.

Reproducibility

To ensure methodological transparency and support further research, the full DuckDB ingestion scripts, transformation pipelines, and hyperparameter configuration matrices utilized in this study are open-source and available at: https://github.com/gentleman-programming/pp-pipe-dt.

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